MIL-B-7525B
29 January 1971
SUPERSEDING
MIL-B-7525A(USAF)
3 October 1957
MIL-B-7526B(USAF)
22 October 1957
MIL-B-7527A(USAF)
22 January 1958
MIL-B-7528B(USAF)
22 Oct 1957

MILITARY SPECIFICATION

BRACKET AND RECEPTACLE, HOT CUP, AIRBORNE

This specification is mandatory for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of hot cup brackets and receptacles to be used aboard aircraft with hot cups conforming to MIL-H-21303.

1.2 Classification. Hot cup brackets and receptacles shall be of the following types as specified (see 6.2).

Type A-1	4-Cup unit, 28 volts (see fig 1 and 2)
Type A-2	1-Cup unit, 28 volts (see fig 5 and 6)
Type B-1	4-Cup unit, 115 volts (see fig 3 and 4)
Type B-2	1-Cup unit, 115 volts (see fig 7 and 8)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250	Aluminum Alloy, Plate and Sheet, General Specification for
QQ-A-250/1	Aluminum Alloy 1100, Plate and Sheet
QQ-A-250/2	Aluminum Alloy 3003, Plate and Sheet
QQ-S-766	Steel Plate, Sheet, and Strip - Corrosion Resisting
QQ-W-423	Wire, Steel, Corrosion Resisting
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging

FSC 7310

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Military

MIL-P-116	Preservation, Methods of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-S-13572	Spring, Helical, Compression and Extension
MIL-H-21303	Hot Cups, Liquid, Electric, Airborne

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS 25231	Lamps, Incandescent, Single Contact, Miniature Bayonet Base
MS 35206	Screw, Machine, Pan-Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNC-2A

Air Force-Navy Aeronautical

AN373	Nut, Self-Locking, Plate, Countersunk, 100°, 250°F

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H-28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

National Motor Freight Traffic Association, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Attn: Tariff Order Section, 1616 P St., N. W., Washington, D.C., 20036.)

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Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Il. 60606.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), this specification provides for first article testing and inspection. A first article comprises a preproduction model or initial production sample; approval involves testing and evaluating the first article for conformance with the requirements herein before, or in the initial stage of production under a contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. Hot cup brackets and receptacles shall be either single or four cup units and shall be designed to conform to figures 1, 3, 5, or 7. Types A-1 and A-2 shall be designed to operate on 28 volts and Types B-1 and B-2 shall be designed to operate on 115 volts. The back of each bracket shall be readily removable for access to the components and wiring. All components shall be mounted in such a manner that will allow easy maintenance or replacement using tools and maintenance equipment normally available as commercial standards. Special and commercial standard tools are defined in MIL-S-8512.

3.2.1 Material. The hot cup bracket shall be constructed of 0.032 inch thick anodized aluminum conforming to QQ-A-250/1 (H14 or H24), or QQ-A-250/2 (H14 or H-24), or corrosion resisting steel conforming to QQ-S-766, class 201, 202, 302, or 304. Corrosion resisting steel shall be 0.024 inch thick (24 gauge) with at least a 180 grit finish.

3.2.2 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement without modification of parts or equipment.

3.2.3 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3 Construction. The hot cup brackets shall be of spot-weld construction, shall be complete and ready for use when connected to a suitable source of power. The brackets shall be fabricated of parts which are sound and free from repair. Defective parts shall not be reclaimed for use.

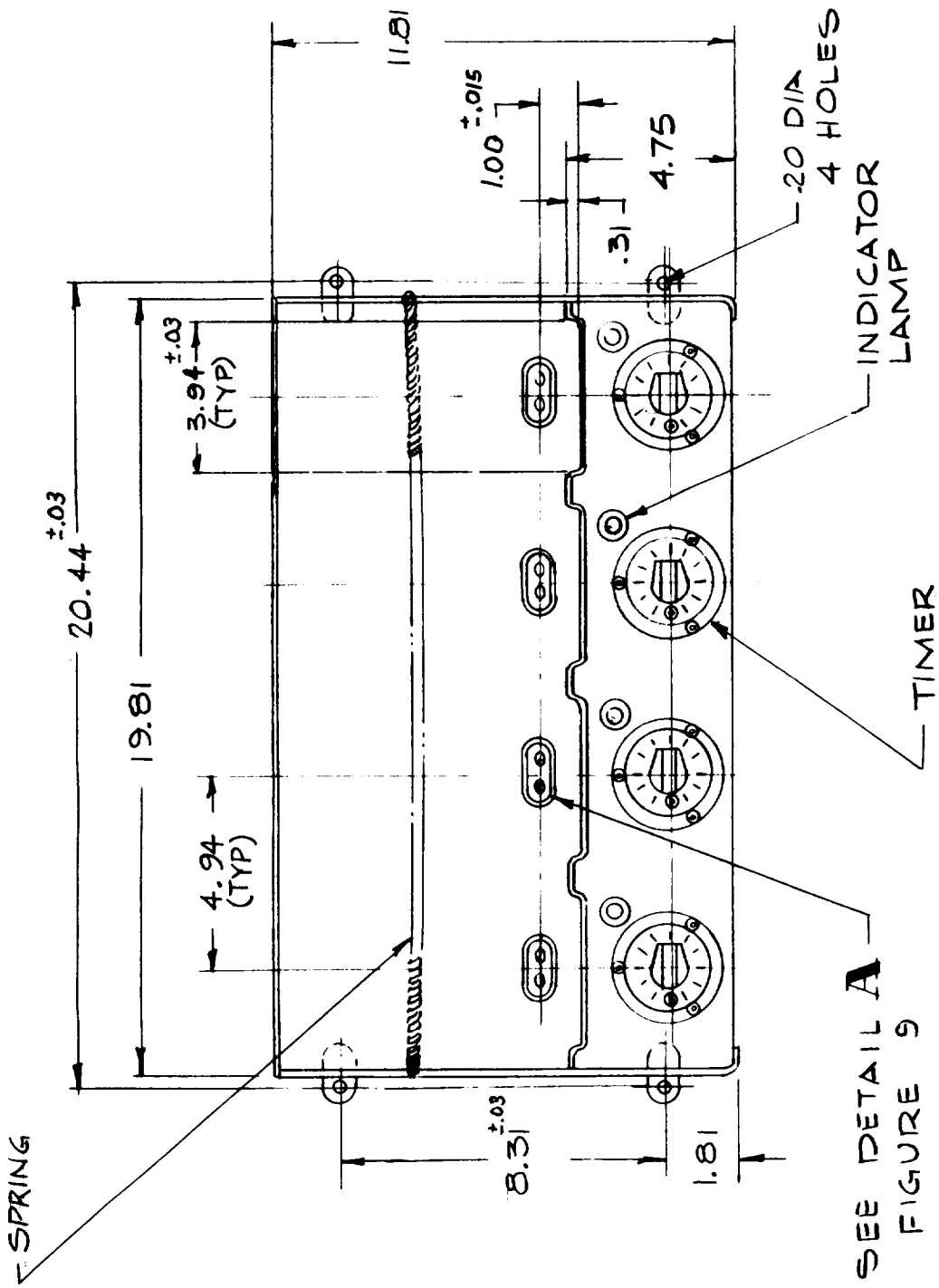
3.3.1 Welding. Spot welding shall be in accordance with MIL-W-6858 for the material being used. All discoloring shall be removed from the weld areas.

3.3.2 Fastening devices. All screws, pins, and similar parts shall be installed with means to prevent loss of tightness. Those subject to removal or adjustment shall not be swaged, peened, or otherwise permanently deformed.

3.4 Components. The hot cup bracket shall consist essentially of a cup retainer, receptacle, indicator light, timer, and electrical cord.

3.4.1 Spring retainer. Each bracket shall be furnished with a 3/16 inch diameter coil extension spring for retaining hot cups conforming to MIL-H-21303

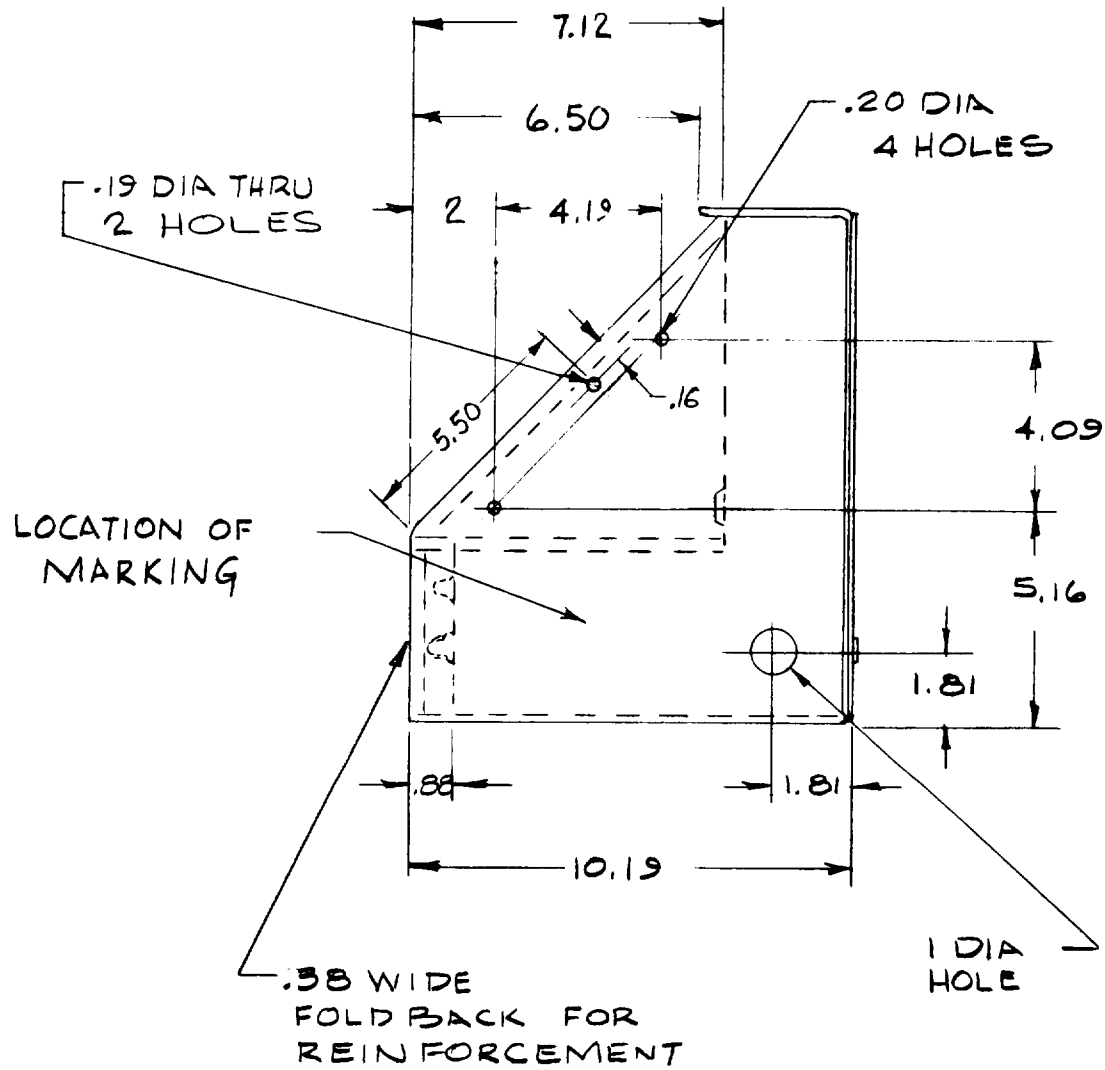
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TYPE A-1

FIGURE 1

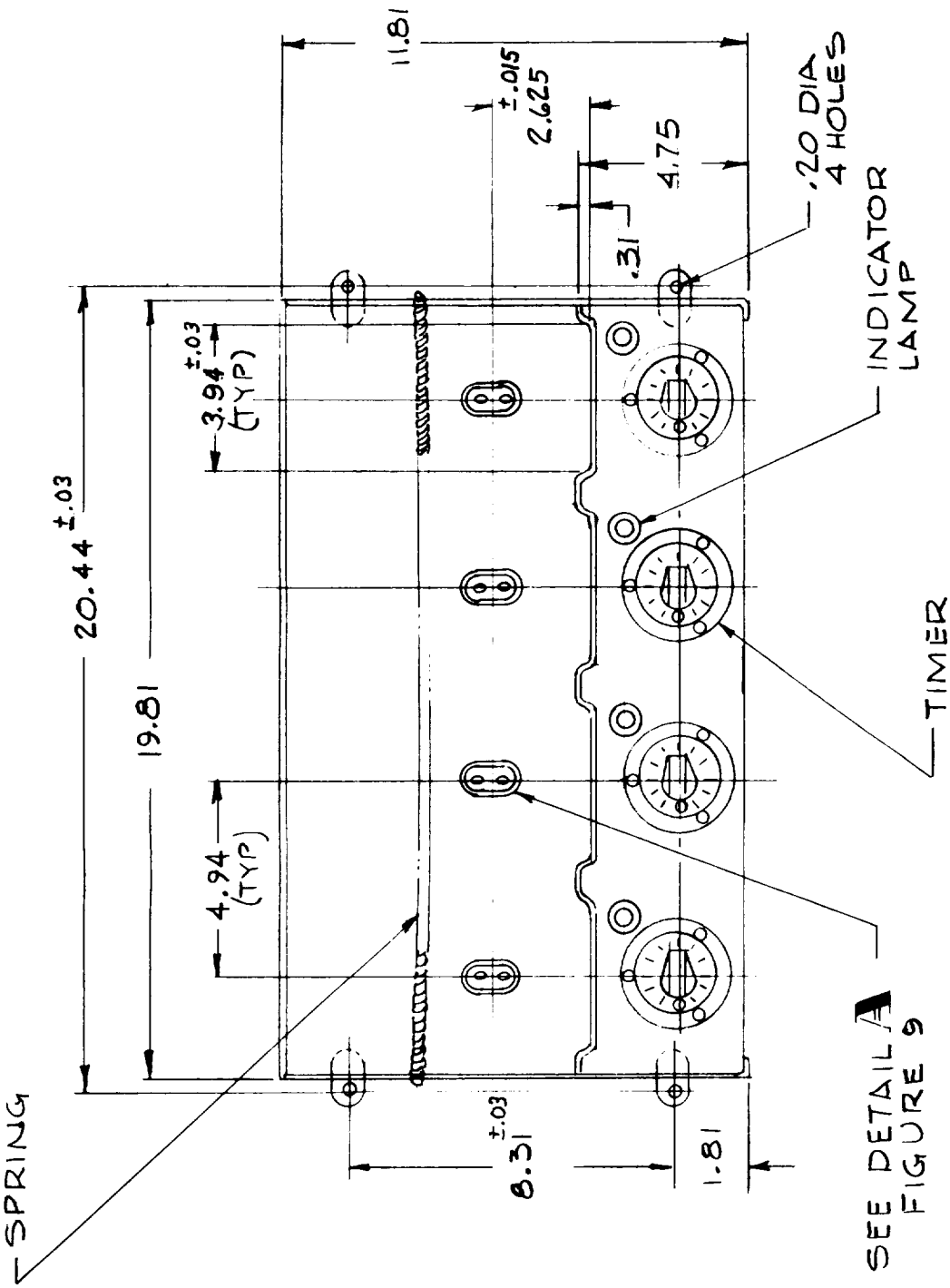
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A-1 BRACKET
SIDE VIEW

FIGURE 2

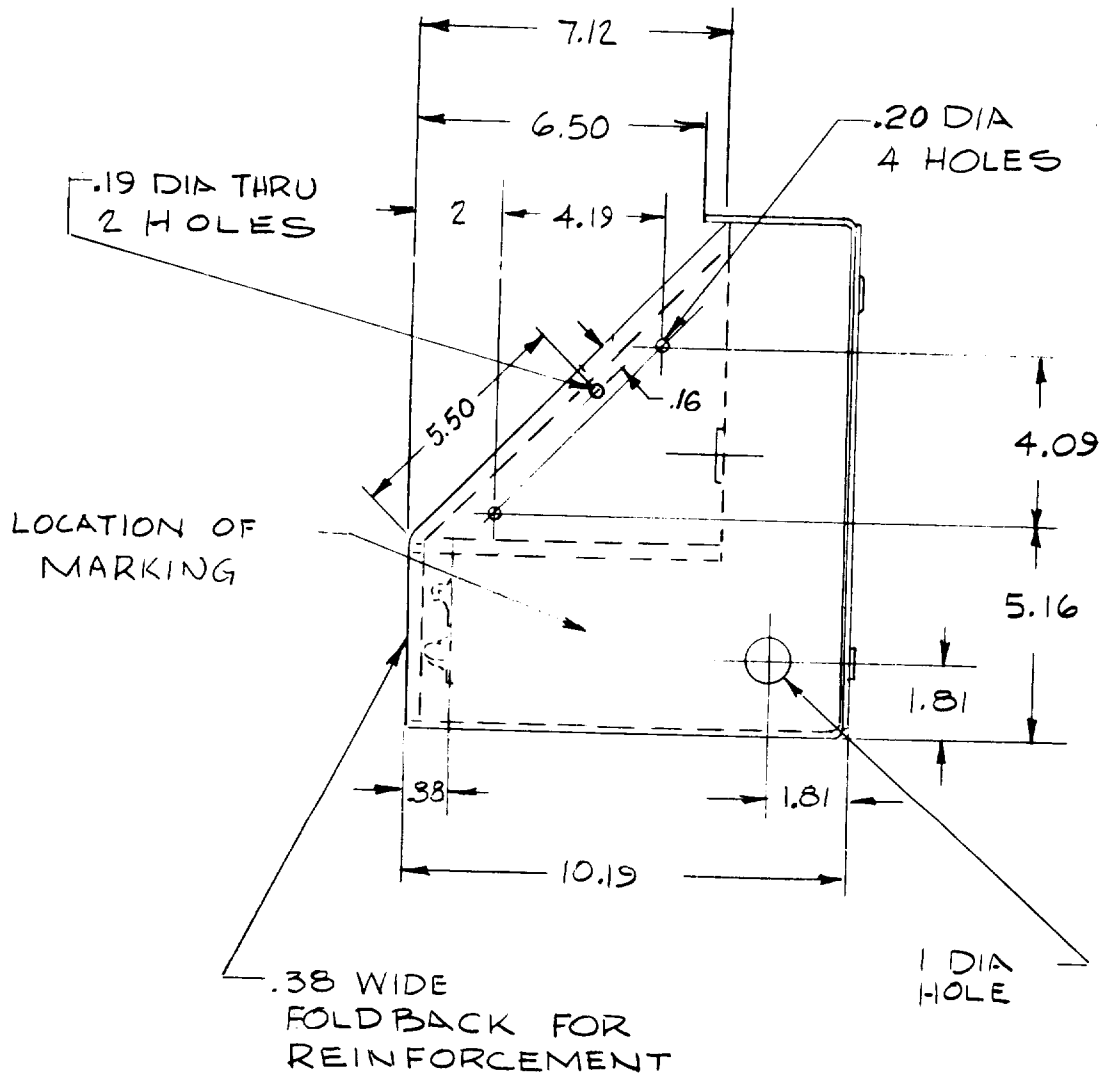
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TYPE B-1

FIGURE 3

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B-1 BRACKET
SIDE VIEW

FIGURE 4

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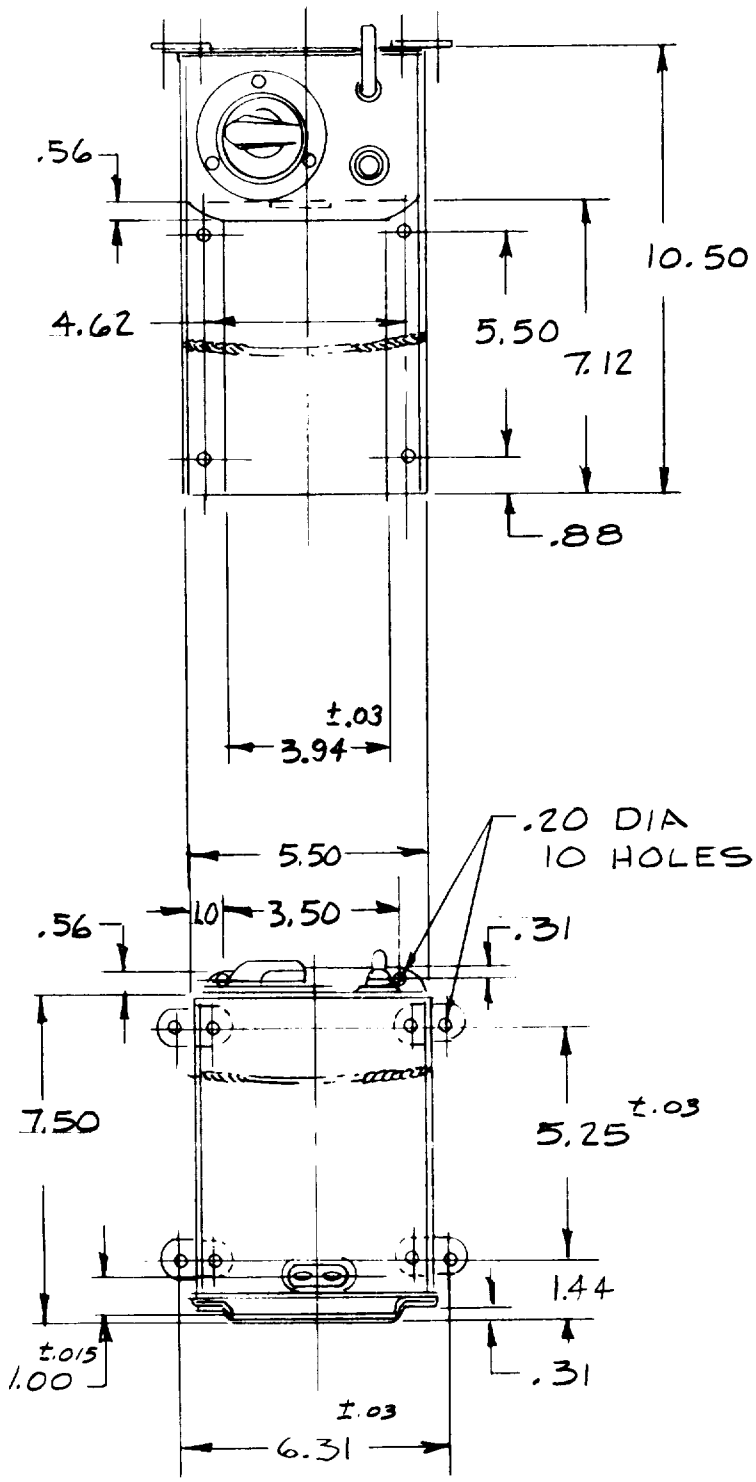
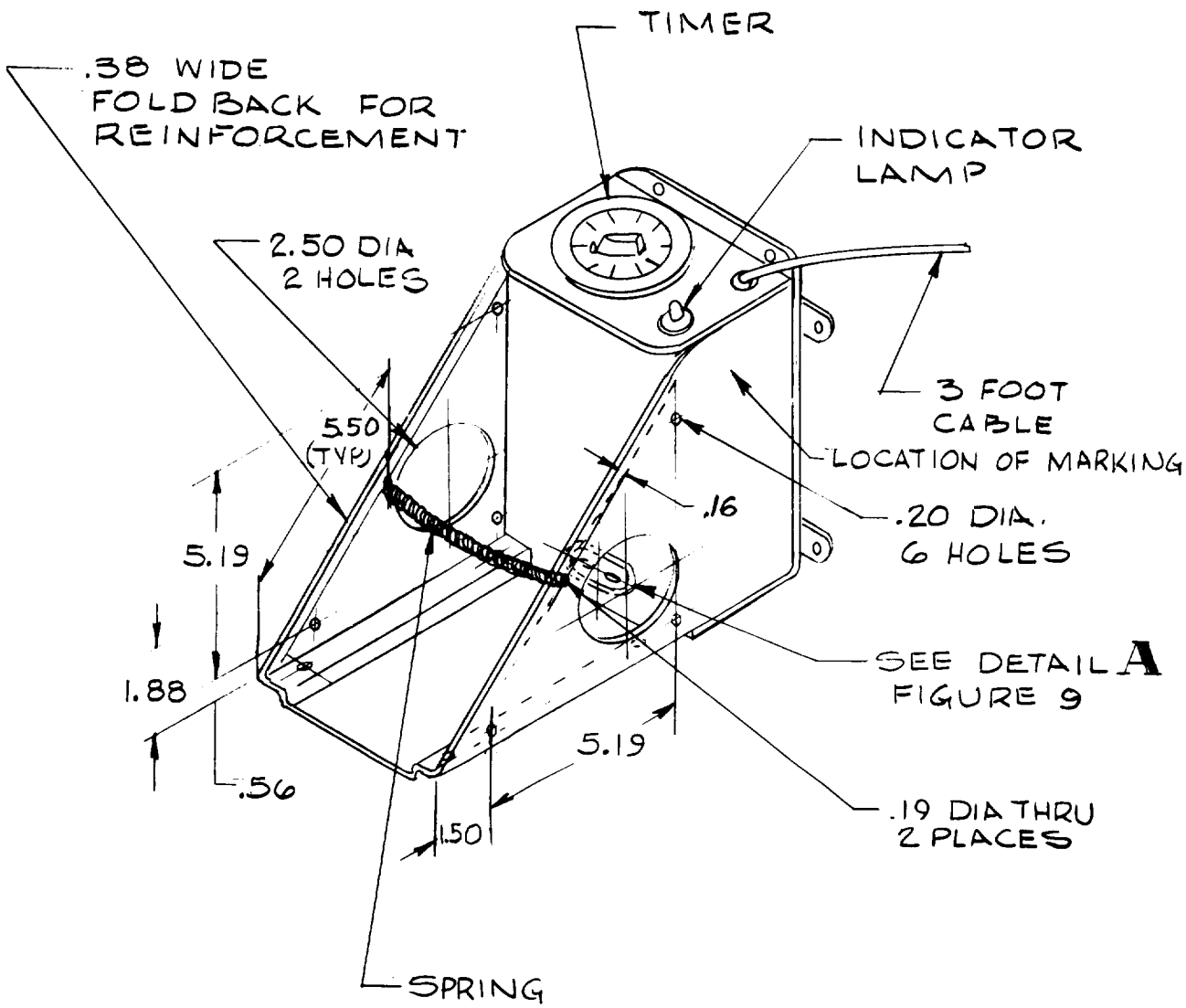


FIGURE 5

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TYPE A-2

FIGURE 6

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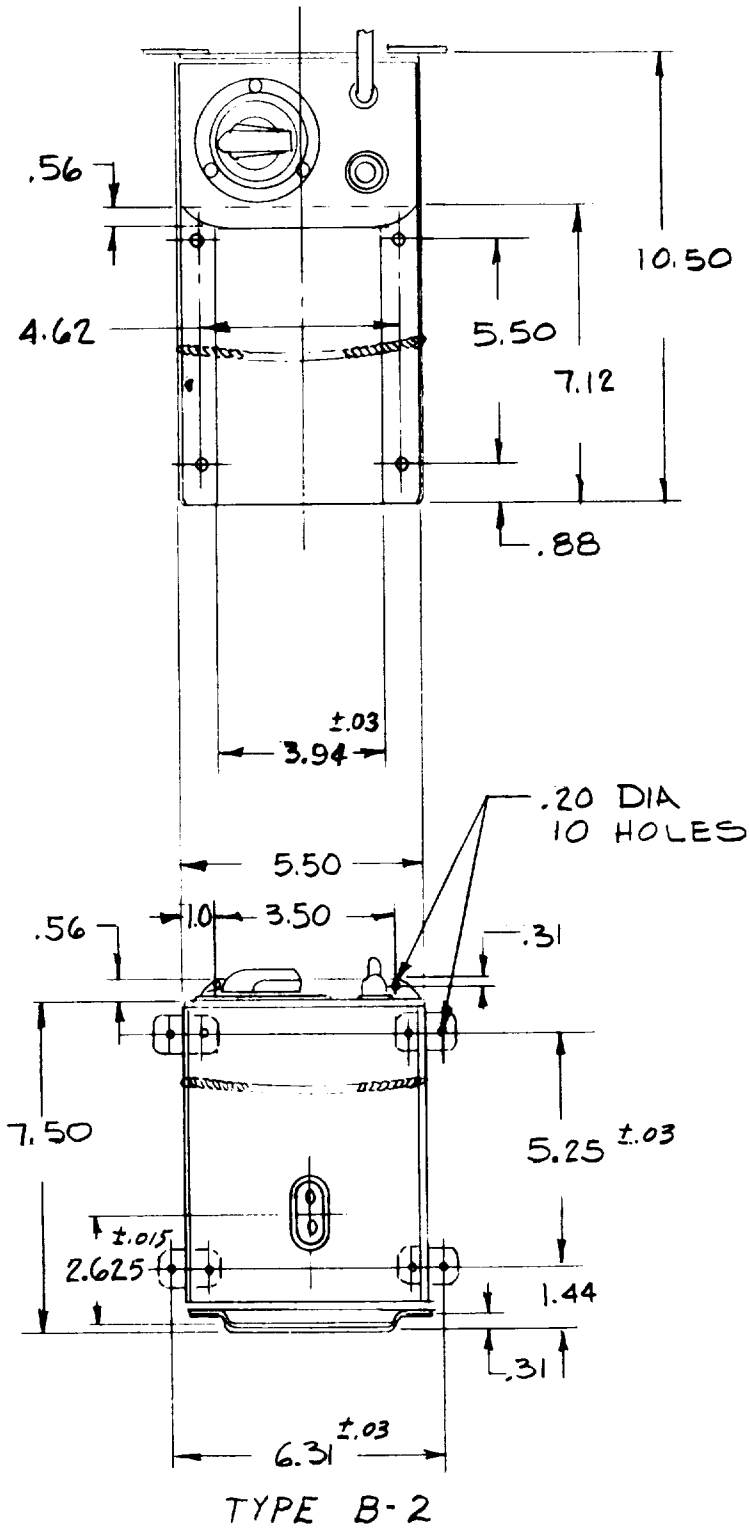
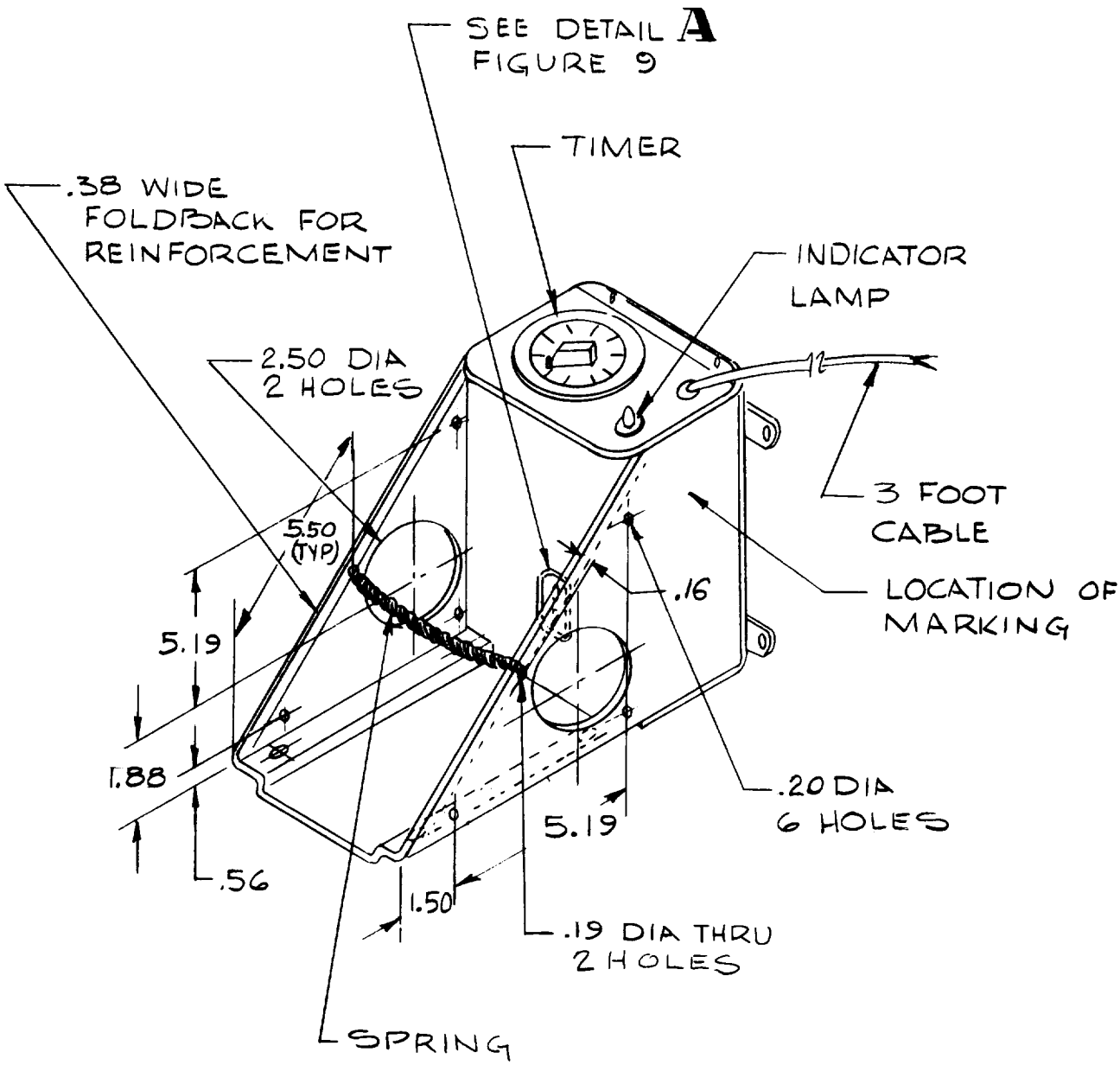


FIGURE 7

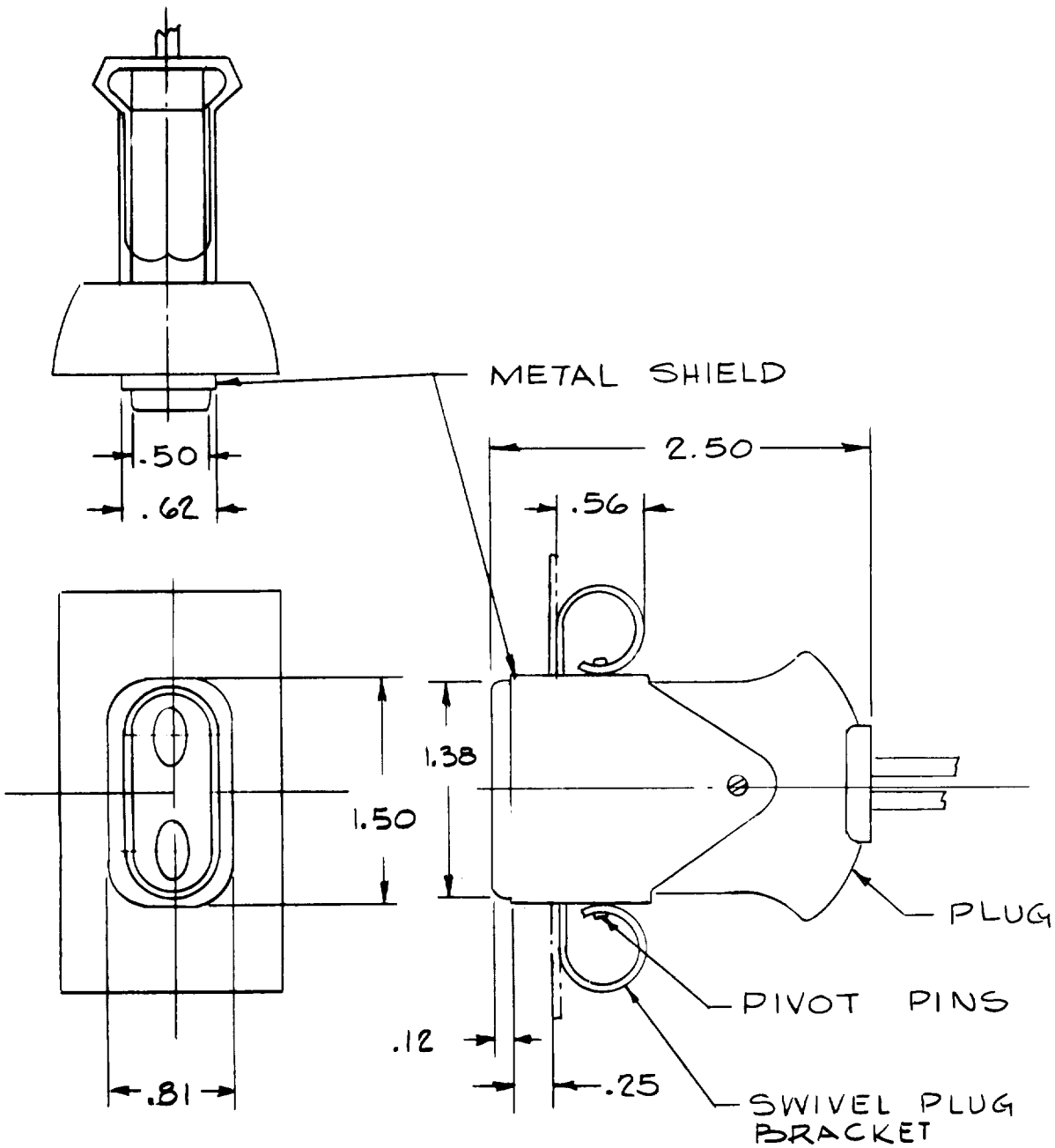
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TYPE B-2

FIGURE 8

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DETAIL A

FIGURE 9

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in position. The spring shall conform to Type II, Grade B of MIL-S-13572 and shall be fabricated of 0.020 minimum diameter corrosion resistant steel wire conforming to QQ-W-423. The spring shall be designed for easy fastening to the bracket when a hot cup is in place. One end shall be permanently attached. The free length for Types A-1 and B-1 shall be 18 inches and the free length for Types A-2 and B-2 shall be 6 inches.

3.4.2 Receptacle(s). Each bracket shall be provided with one or four (as applicable) receptacle(s) for plugging in hot cups conforming to MIL-H-21303. The receptacles shall be in accordance with figure 9 and shall be mounted on pivots to allow sideways movement. Each receptacle shall have a steel, wrap-around jacket with studs for mounting into the swivels. The swivels shall allow a total sideways motion of approximately 15 degrees. The contact pin holes in the receptacles of Types A-1 and A-2 brackets shall make satisfactory electrical contact with Type I hot cups conforming to MIL-H-21303. The contact pin holes in the receptacles of Types B-1 and B-2 brackets shall make satisfactory electrical contact with Type II hot cups conforming to MIL-H-21303.

3.4.3 Indicator light(s). Each bracket shall be provided with one or four (as applicable) indicator light(s) to indicate when power is being applied to each hot cup. Each light shall have an amber, transparent, removable plastic cover to protect the bulb. Type A-1 and A-2 lights shall operate directly from the 28 volt power source and Type B-1 and B-2 lights shall operate directly from the 115 volt power source. Each light shall be wired with a timer to come on when the current is applied and to go off when the timer cuts off the current. Bulbs for Type A-1 and A-2 shall be in accordance with MS-25231-1819.

3.4.4 Timer(s). Each bracket shall be provided with one or four (as applicable) mechanical timer(s) with snap action switch, constructed and installed so that power may be applied to each hot cup for any preset length of time from 0 to 30 minutes. Each timer shall be so constructed that, after setting, movement of the knob in either direction will not damage the timer. An internal stop shall be provided to prevent the timer knob from turning more than one complete revolution. The timer mechanism shall require no activation other than setting, except that initial cocking shall be permissible. The timer scale shall be graduated in minutes and the number of minutes shall be indicated every five minutes. The timers for Types A-1 and A-2 brackets shall be rated at 22 amperes at 28 volts direct current. The timers for Types B-1 and B-2 brackets shall be rated at six amperes at 115 volts alternating current. Each timer shall have an accuracy of plus or minus ten percent per hour over a temperature range of zero degrees to ninety degrees Fahrenheit.

3.4.5 Electrical cord. Each hot cup bracket shall be provided with a 3-foot, heavy duty, rubber covered cord for the power source. The cord shall be sized to carry the current specified in 3.6.

3.5 Mounting provisions. Each bracket shall be provided with four removable lugs for installation. The bracket body, bracket back plate and lug will be assembled with self-locking plate nut AN 373F832A and pan head machine screw MS 35206-245 or equal. The brackets shall be furnished with the lugs attached to the brackets.

3.6 Electrical. All wiring and electrical connections shall be made in accordance with the applicable portions of MIL-W-5088. Wiring for Type A-1 brackets shall be designed for a total maximum direct current of 88 amperes at 28 volts and wiring for Type A-2 brackets shall be designed for a total maximum direct current of 22 amperes at 28 volts. Wiring for Type B-1 brackets shall

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be designed for a total maximum current (both AC and DC) of 24 amperes at 115 volts and wiring for Type B-2 shall be designed for a total maximum current (both AC and DC) of 6 amperes at 115 volts. Type A-1 and B-1 brackets may utilize insulated copper bus bars inside rear of each unit where each station may be wired to.

3.7 Performance. Each hot cup bracket shall meet the following performance criteria.

3.7.1 Operation. Each hot cup bracket, when connected to a suitable power source (28 or 115 volts as applicable), and with a voltmeter connected to the receptacle, shall pass the tests specified in 4.5.1.

3.7.2 Timer accuracy. Timers shall operate within the accuracy requirement of 3.4.4 when tested in accordance with 4.5.2.

3.8 Identification. Each bracket shall be permanently marked for identification in accordance with MIL-STD-130. Identification shall include the following information:

Nomenclature
Electrical Characteristics
Manufacturer's Name
Manufacturer's Identification
Contract or Order Number
Federal Stock Number
Date of Manufacture
U S

3.9 Workmanship. The hot cup brackets shall be constructed in accordance with good commercial practices. Each bracket shall be uniform in quality and temper, clean, sound, smooth, commercially straight or flat, and free from injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass the examination or any test shall be cause for disapproval of the first article.

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4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on sample units selected in accordance with 4.3. Unless otherwise specified (see 6.2), quality conformance inspection shall comprise the examination in 4.4 and the tests in 4.5. Failure of any unit to pass the examination or any test shall be cause for rejection.

4.3 Sampling. Random samples of hot cup brackets shall be selected for examination and tests from each lot in accordance with inspection level II of MIL-STD-105. For sampling purposes, a lot shall consist of all hot cup brackets of the same type being offered for delivery to the Government at one time under a specific contract. A sample unit shall consist of one completely fabricated hot cup bracket.

4.4 Examination. Each hot cup bracket and receptacle shall be examined to determine compliance with the requirements of this specification with respect to design, general construction, and workmanship,

4.5 Tests. All tests required by this specification shall be made at atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25 degrees Centigrade (77 degrees Fahrenheit). Where tests are made with atmospheric conditions substantially different from these values, proper allowance shall be made for the change in instrument readings. Sampling for all tests shall be as specified in 4.3.

4.5.1 Operation test. Each hot cup bracket sample shall be connected to a suitable power source of 28 volts or 115 volts as applicable. A suitable voltmeter shall be connected to each receptacle. One or all four timers (as applicable) shall be turned to the 30-minute setting and then returned manually to the "OFF" position. The timer shall then be returned to the 30-minute setting and allowed to run to the "OFF" position. Indicator lights shall be observed for proper functioning and the voltmeter shall be observed to determine that full voltage is available at the receptacle at all times that the timer is not in the "OFF" position.

4.5.2 Timer accuracy test. The accuracy of each timer shall be checked at a temperature of zero degrees Fahrenheit by testing the full 30-minute settings with a stop watch. Each timer shall meet the accuracy specified in 3.4.4.

4.5.3 Weight and dimension test. Each bracket sample shall be weighed to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each bracket shall be preserved and packaged in accordance with method III of MIL-P-116 and 5.3. Unit containers shall be weather resistant type of PPP-B-636. Water-proofing of openings of slotted style boxes is required when single units are shipped overseas without being overpacked into shipping containers in accordance with 5.2. .

5.1.2 Level C. Brackets shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Brackets preserved and packaged in accordance with 5.1.1 shall be packed in overseas type shipping containers meeting PPP-B-601. When single units are shipped overseas, the unit containers shall be accepted as the shipping containers. Containers shall contain identical quantities, shall be of uniform shape and size, and shall be of minimum cube and tare consistent with protection required. Containers shall be provided waterproof protection in accordance with 5.3, and shall be closed and strapped in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Brackets preserved and packaged in accordance with 5.1.1 shall be packed in domestic type shipping containers meeting PPP-B-636. Containers shall contain identical quantities, shall be uniform in shape and size, and shall be of minimum cube and tare consistent with protection required.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection. Unless otherwise specified (see 6.2), cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special markings required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hot cup brackets covered by this specification are intended for installation in aircraft for use with electrically heated hot cups conforming to MIL-H-21303.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type hot cup bracket required (see 1.2).
c. First article, if not required (see 3.1).
d. First article inspection, if different (see 4.2.1).
e. Quality conformance inspection, if different (see 4.2.2).
f. Levels of preservation, packaging and packing (see 5.1 and 5.2).
g. Cushioning, blocking, bracing, if different (see 5.3).

6.3 Supersession data. This specification combines and updates the requirements of MIL-B-7525A(USAF) dated 3 October 1957 for Type B-2, MIL-B-7526B(USAF) dated 22 October 1957 for Type A-2, MIL-B-7527A(USAF) dated 22 January 1958 for Type B-1, and MIL-B-7528B(USAF) dated 22 October 1957 for Type A-1. Type designations have not been changed.

Custodians:
Army - GL
Navy - AS
Air Force - 84

Review activities:
Army - GL
Navy - AS, YD

Preparing activity:
Air Force - 84

Project number:
7310-0343

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. 7 AFLC RAFB, GA JUN 67 - 500

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